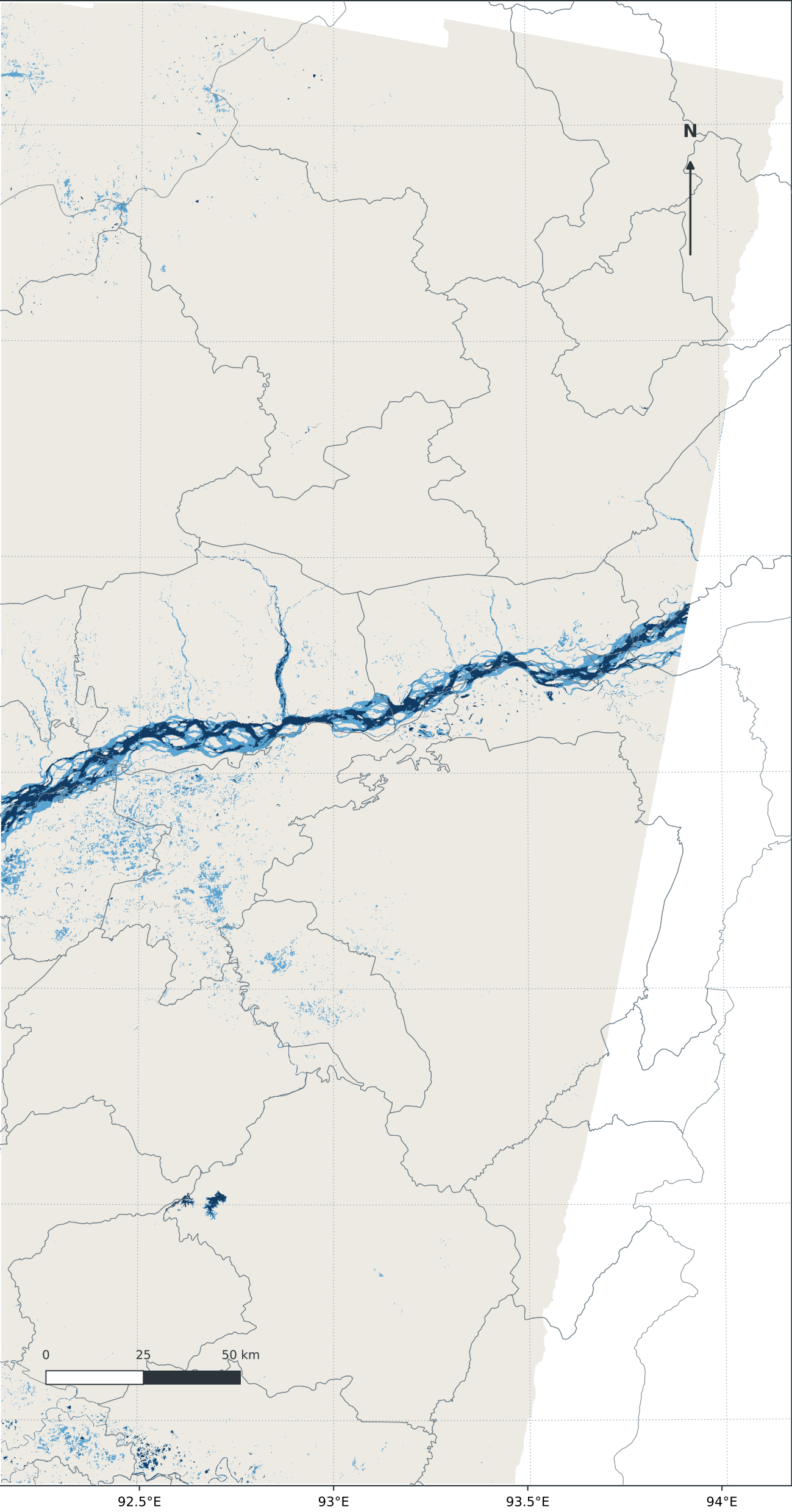


Brahmaputra valley flood extent and flooded cropland

Sentinel-1 SAR, 12 August 2016 · monsoon season, Kharif 2016 · central Brahmaputra valley, Assam and adjoining states

Flood extent, 12 August 2016



LEGEND

- Permanent water

JRC occurrence $\geq$ 50%
- Flood water

dark on 12 Aug, not permanent
- Land, imaged
- No data

outside swath, or edge buffer

AREA BY CLASS

class	hectares	operating-point range
Flood water	119,779	119,779 to 170,625
Flooded cropland	46,943	46,943 to 87,127
Permanent water	58,389	58,389 to 60,076
Land, imaged	6,034,074	6,034,074 to 6,086,607
No data	1,484,816	—

Range spans two defensible thresholds: -18.75 dB, maximum agreement with hand labels, and -17.75 dB, mapped area unbiased against them. Not a confidence interval.



PROJECTION	EPSG:32646 (UTM zone 46N), 20 m pixel
SENSOR	Sentinel-1A IW GRD, RTC gamma0, VH polarisation
ACQUISITION	12 August 2016, 23:46 UTC, relative orbit 77, descending
METHOD	Lee 5×5 speckle filter in linear power, single-date threshold, 600 m swath-edge buffer to no data, slope > 8° reclassified as land, minimum mapping unit 0.4 ha
PERMANENT WATER	JRC Global Surface Water occurrence $\geq$ 50% (Pekel et al. 2016)
CROPLAND	ESA WorldCover 2021 class 40. Four to five years after the event; no open 10 m cropland map exists for 2016
BOUNDARIES	geoBoundaries ADM2 India, current vintage, CC BY 4.0. Assam districts have been created since 2016
VALIDATION	Sen1Floods11 hand labels, 65 chips, IoU 0.519 (95% CI [0.395, 0.609]). The refinement steps above are unvalidated: no labelled data covers them
PRODUCED	10 September 2026 · github.com/sruthi-swathandran/geospatial-portfolio